AOTN CHALLENGE · TSYP 14 · PHASE 2.1

Solution Blueprint

Team name: The Architects **Student Branch:** IEEE ENSI SB **Solution name:?????**

_____

How to use this template: make a copy (File > Make a copy), answer the guiding questions under each heading, then delete the questions in grey. Max 6 pages. Deadline: 20 October 2026, 11:59pm. Submit the link through the Phase 2.1 form.

1. The Solution

What is your solution, in one sentence? (choose one)

- Rassed is an innovative, non-intrusive remote-monitoring and alerting solution that uses computer vision and audio analysis to monitor existing gauges, detect alarm signals, predict abnormal trends, and alert maintenance teams via SMS or voice calls, improving the safety and reliability of MT/BT transformer substations in Tunisian factories without modifying the existing equipment.
- Rassed is a smart remote-monitoring solution that uses a camera and microphone to monitor MT/BT transformer substations in Tunisian factories and alert maintenance teams to potential risks without modifying existing equipment.

Who is it for (your main user)?

The main user is the on-call qualified electrician (habilité) responsible for the substation. Also the night or weekend security guard, who is on site but not qualified to enter the room, and the plant manager, who follows the dashboard

What are its 3 to 5 key features? (fema alarm lamps?)

- **Real-Time Monitoring:** A camera reads existing gauges and alarm lamps, while a microphone detects the siren, providing live monitoring through a dashboard.

- **Predictive Maintenance:** Analyzes historical readings to detect abnormal trends and anticipate potential failures before they occur.

- **Smart Alerting:** Sends instant SMS and push notifications to the electrician and night guard, with alerts classified as WATCH or INTERVENE NOW.

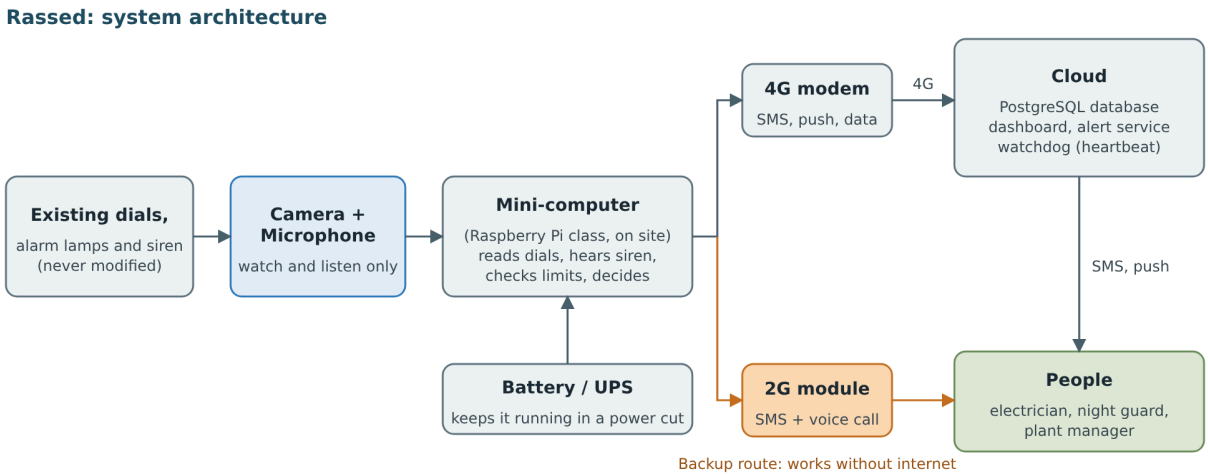
- **Backup Communication System:** Uses 4G as the primary communication channel, with 2G SMS and automated voice calls as backup if 4G fails.

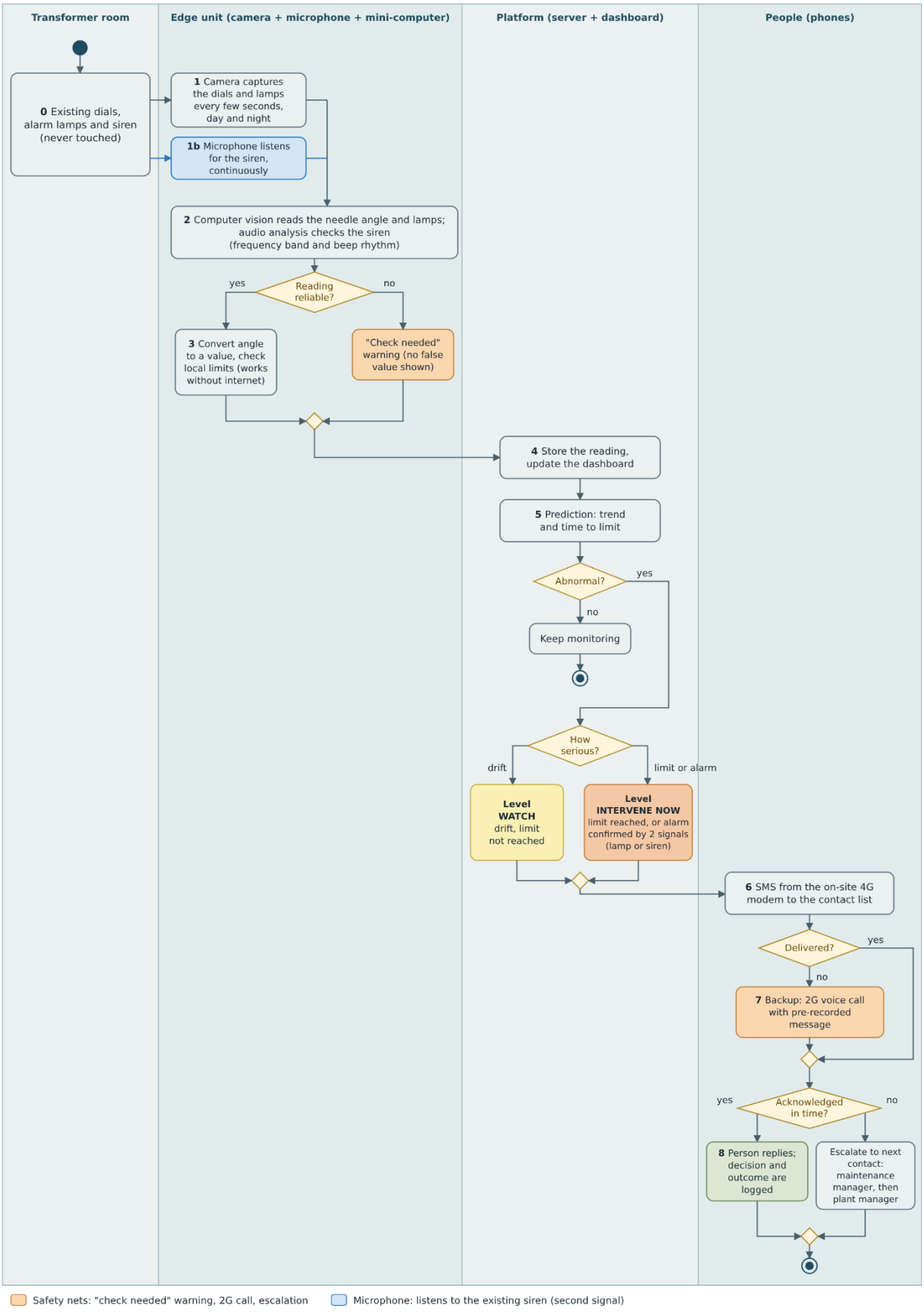
- **Decision Log:** Records every alert and human response, including who was notified, when it was acknowledged, the decision taken, and the outcome, ensuring traceability and continuous improvement.

2. How It Works (choose one diagram)

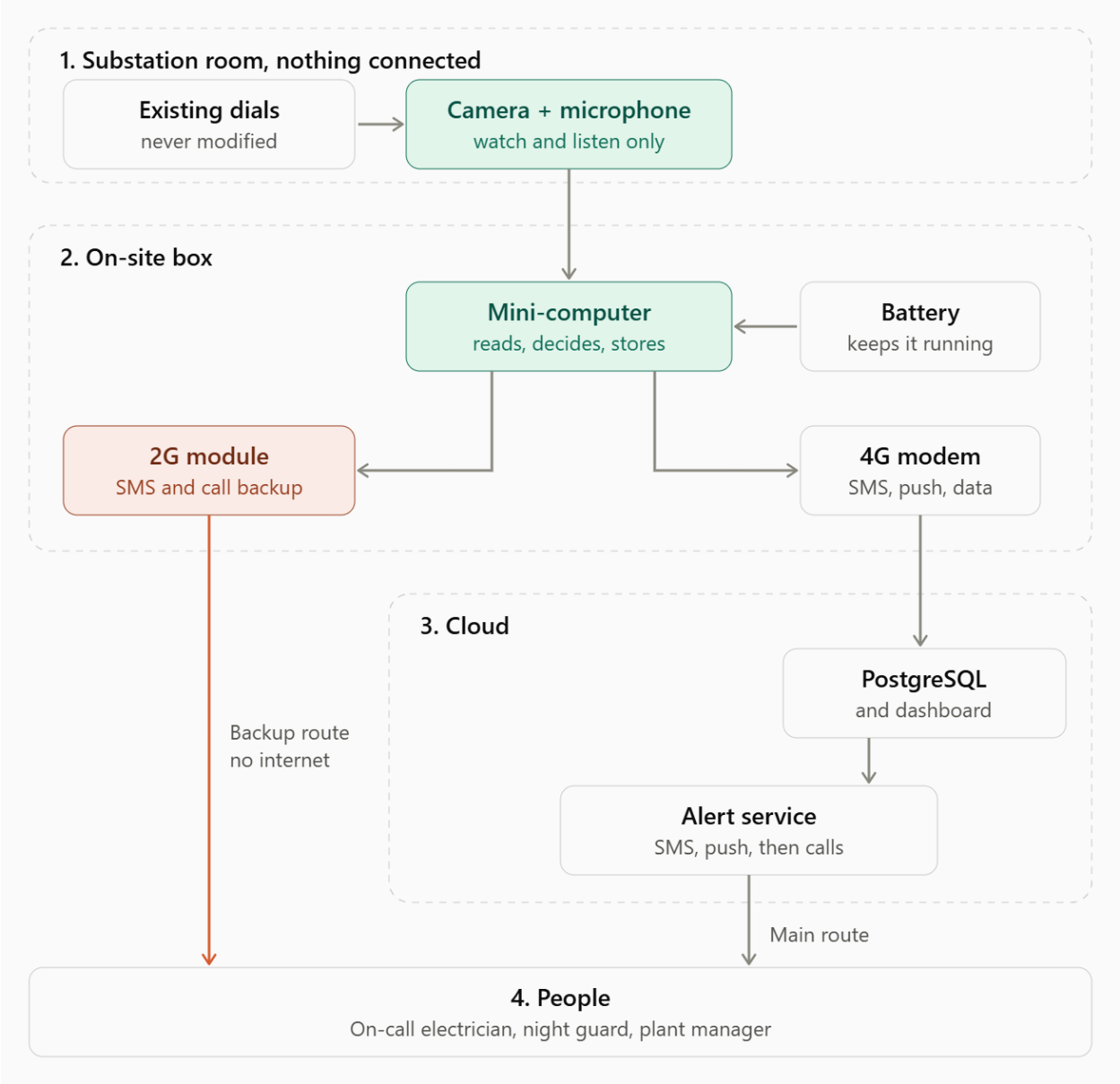
Add one simple diagram: a user journey (step by step, what the user does) or a system architecture (the main building blocks and how they connect). Explain it in a few lines.

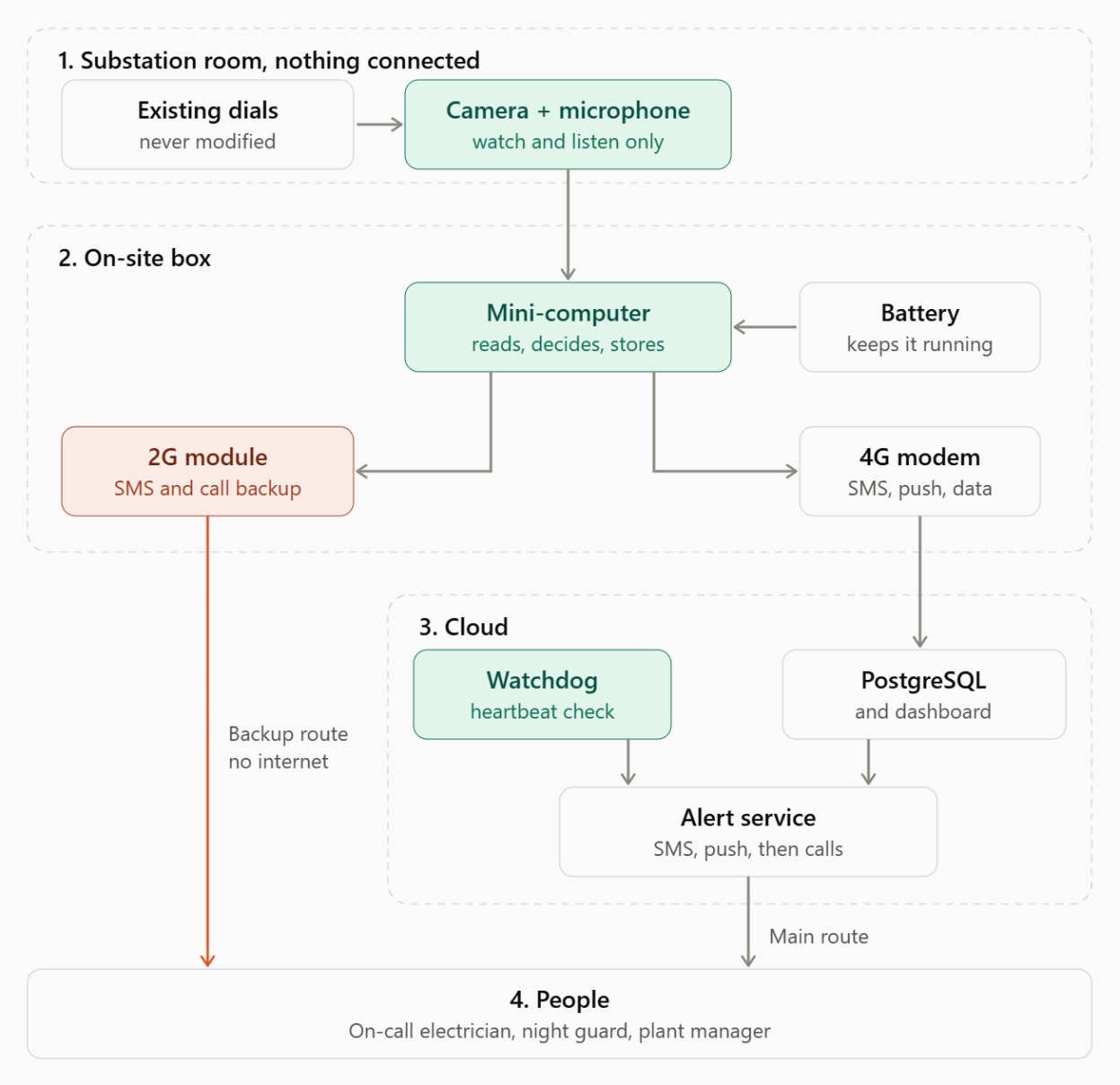
Diagram




Transformer room
Edge unit (camera + microphone + mini-computer)
Platform (server + dashboard)
People (phones)
0 Existing dials, alarm lamps and siren (never touched)
1 Camera captures the dials and lamps every few seconds, day and night
1b Microphone listens for the siren, continuously
2 Computer vision reads the needle angle and lamps; audio analysis checks the siren (frequency band and beep rhythm)
Reading reliable?
yes
no
3 Convert angle to a value, check local limits (works without internet)
"Check needed" warning (no false value shown)
4 Store the reading, update the dashboard
5 Prediction: trend and time to limit
Abnormal?
yes
no
Keep monitoring
How serious?
drift
limit or alarm
Level WATCH drift, limit not reached
Level INTERVENE NOW limit reached, or alarm confirmed by 2 signals (lamp or siren)
6 SMS from the on-site 4G modem to the contact list
Delivered?
yes
no
7 Backup: 2G voice call with pre-recorded message
Acknowledged in time?
yes
no
8 Person replies; decision and outcome are logged
Escalate to next contact: maintenance manager, then plant manager
Safety nets: "check needed" warning, 2G call, escalation
Microphone: listens to the existing siren (second signal)

system architecture





(Watchdog cloud : indispensable

C’est le seul qui prévient les humains quand la boîte ne peut plus rien envoyer (coupure de courant, batterie vide, boîtier mort, perte totale de réseau). Sans lui, une panne de Rassed passe pour une situation normale, et c’est exactement le risque n°3 de ton rapport.

Watchdog matériel : utile mais secondaire

Il redémarre automatiquement la boîte si son logiciel se fige. Sans lui, un blocage du logiciel déclenche quand même l’alerte SYSTEM OFFLINE via le watchdog cloud, mais quelqu’un doit se déplacer pour redémarrer la boîte. Avec lui, la boîte se répare seule dans la plupart des cas, avant même que l’alerte ne parte. Sur un Raspberry Pi, cette fonction est déjà intégrée et

s’active par une simple configuration, donc le coût est quasi nul.) nzidouh ou pas ?

(choose explanation or step by step)

Explanation

Rassed fixes a camera and a microphone next to the transformer, connected to a small on-site computer (Raspberry Pi class). This computer reads the dials, hears the siren, checks the limits and predicts drifts, even without internet. Readings are sent over 4G to a cloud platform (PostgreSQL database and dashboard), which sends SMS and push alerts to the electrician and the guard. If 4G is unavailable, the 2G module sends the SMS and places a voice call with a pre-recorded message. Nothing is connected to the transformer.

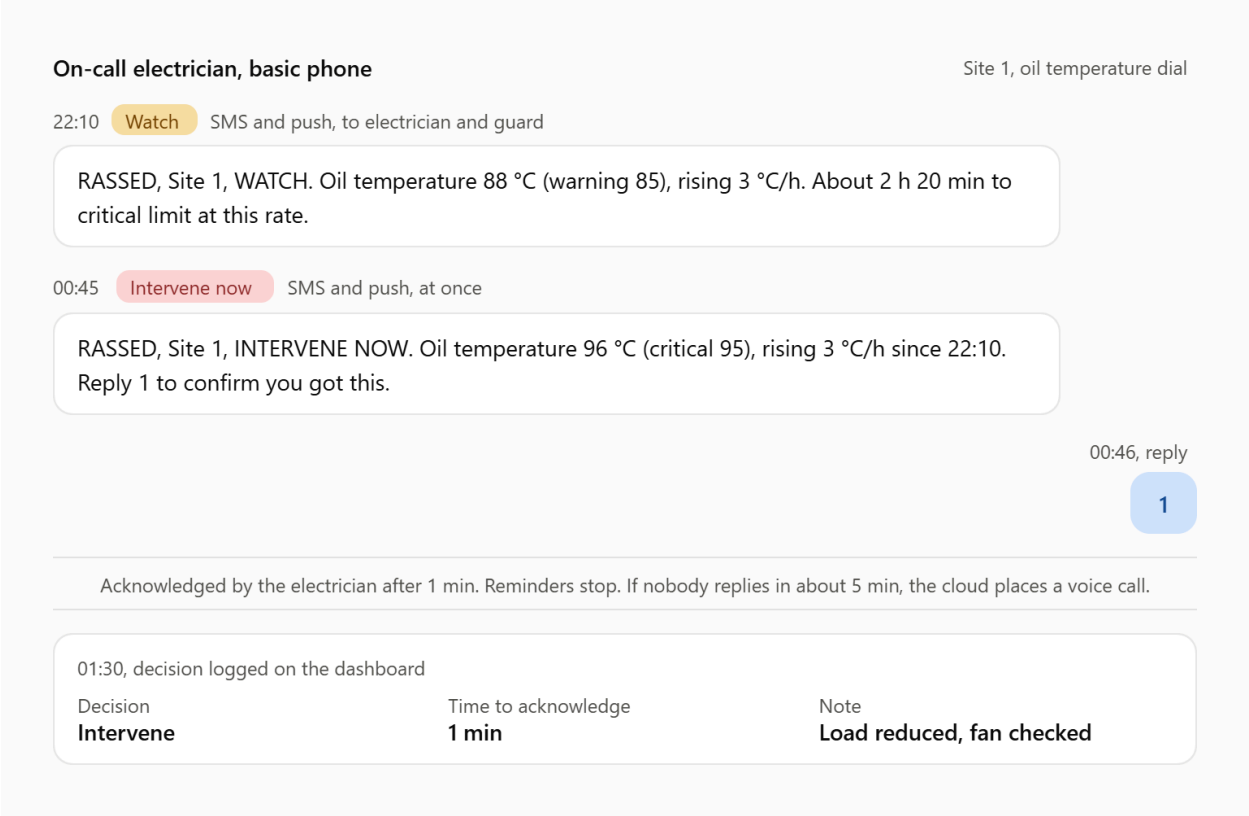
Step by step, from dial to phone

#	Step		What happens
1	Data Capture	Camera & Microphone	The camera observes the existing dials and alarm lamps, while the microphone listens for the siren.
2	Local Processing	Raspberry Pi + Python/Open CV	The system reads the dial values, detects alarm lamps, and identifies the siren.
3	Risk Detection	Limits & Trend Analysis	Rassed checks the readings against predefined limits and detects abnormal trends.
4	Alert Validation	Signal Verification	For a critical alert, Rassed checks whether at least two signals agree: the dial value, alarm lamp, or siren.
5	Data Transmission	4G / 2G Backup	Readings and alerts are sent to the cloud via 4G. If 4G fails, the 2G module provides backup SMS and voice calls.
6	Cloud & Dashboard	PostgreSQL + Web Dashboard	The platform stores readings, alerts, decisions, and history, and displays monitoring data.
7	Alert Notification	SMS, Push Notification & Voice Call	The electrician and night guard receive alerts classified as **WATCH** or **INTERVENE NOW**.
8	Human Response	Electrician / Guard	The notified person acknowledges the alert and decides whether to intervene, continue monitoring, or classify it as a false alarm.

9	Decision Log	Database	Rassed records who was alerted, the acknowledgment, the decision, and the outcome.

Key benefit: Existing local alarms (dials, lamps and siren) become a connected monitoring system that can reach the maintenance team even when the factory is unattended, without any change to the certified installation.

Example: the two alert levels and how the electrician responds (illustrative values)



3. Feasibility

Which tools and technologies will you use? Which skills does your team already have, and which are missing? What resources (budget, hardware, data, partners) do you need?

Tools and Technologies

- **Camera and microphone:** IP camera with infrared night vision and a small USB microphone, fixed outside the danger zone. They only watch and listen, and no sensor

is added.

- **Edge device:** Raspberry Pi class mini-computer, so reading and alerts keep working without internet.
- **Computer vision:** Python and OpenCV to find each dial, measure the needle angle and convert it to a value; alarm lamps are read by colour and brightness.
- **Siren detection:** Python (sound spectrum analysis) to detect the siren's frequency band and beep rhythm, calibrated on site.
- **Communication:** 4G modem for data, SMS and push notifications; 2G GSM module for backup SMS and voice calls.
- **Database and cloud:** PostgreSQL for readings, alerts, decisions and history, and a web dashboard.
- **Prediction:** limits and trend first; anomaly detection and machine learning after history is collected.
- **Notifications:** SMS, push notifications and automated voice calls.

Team skills

Already have	Need to develop
Programming and software development	Industrial electrical systems (MT/BT substations)
Databases and dashboard development	Embedded / IoT: Raspberry Pi, 4G and GSM modems
Data analysis and machine learning	Computer vision for dial reading
System design and problem-solving	Audio signal processing for siren detection; industrial cybersecurity

Required Resources

Resource	Requirements
Budget	Prototype components (camera, microphone, mini-computer, 4G and 2G modems, battery), SIM cards and SMS costs
Hardware	IP camera, USB microphone, Raspberry Pi class mini-computer, 4G modem, 2G GSM module, battery/UPS

Data	Photos and videos of real transformer dials, recordings of the real siren and the background noise, alarm logs and maintenance records
Software	Python, OpenCV, PostgreSQL, a dashboard framework, and machine learning libraries
Partners	Tunisian factories, qualified electricians, and telecom operators (to confirm 2G coverage)

4. Risks

What are the top 3 things that could stop your project? For each one: how likely is it, and what will you do about it?

Risk	Likelihood (low / medium / high)	How we handle it
1.Gauge reading errors	High	Fixed mount with infrared light, one-time calibration, a confidence score on each reading, and a “check needed” message instead of a false value. Train on photos from varied conditions.
2.Alarm sound misdetected	Medium	Record the actual siren at each site, calibrate the threshold per site, and require a match on frequency band and beep rhythm. Combine with the camera so two signals confirm each other.
3.Device failure or going offline	Medium	Battery/UPS backup, 2G communication fallback, local data storage, and a heartbeat mechanism that alerts the team if the system stops responding.

5. Link to Phase 1

Recall your Phase 1 problem in one sentence. Which validated need (from your research and interviews) does each part of your solution answer?

Transformer substations in Tunisian factories already have sensors and alarms, but the warning stays trapped in a room nobody is watching, so at night, on weekends or during shutdowns a slow, catchable drift can grow unseen into a threat to the plant and the people around it.

Validated need (Phase 1)	Part of the solution that answers it
Staff have no remote view of the substation's condition, and abnormal trends go unnoticed until a failure happens	Remote monitoring: the camera reads the dials and the dashboard shows live values and history, with trend detection and the time left before the limit
Technicians and guards are not warned in time when a fault occurs while the factory is unattended, because the local alarm is only audible inside the room	Smart alerting: the camera sees the alarm lamp, the microphone hears the siren, and SMS and push notifications reach the electrician and the guard at once
The warning must still reach someone when the main channel fails, since a missed alert means damage and downtime	Backup communication: 4G as the main channel, with a 2G SMS and voice call with a pre-recorded message as fallback, and a heartbeat
The sensors and gauges are already installed, but the alarm stays local and passive: escalation is missing, not detection	Touch-nothing design: no wire and no new sensor, the camera and the microphone only observe, so there is nothing to re-certify